

The judges scoring, note	Team number: 20025	The judges scoring, note
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Title: Research about recommending books based on Hierarchical analysis method and BP neural network

Abstract

At present, with the development of the information technologies and the Internet, evaluation and ,the recommendation of all kinds of information is increasingly concerned.

As far as the first question, the original data is filtered first. Then we can get Users' evaluation on the books for 5 points. Considering the effect of different factors on the books' score, then the label, social friends, books browsing amount of three groups of data were analyzed by bivariate correlation analysis respectively, so that we can get the number of users on the books, scores and label users' good friends, the history of the book number of pageviews a positive correlation. Impact of users to book score were history views > the number of users' good friends > tag number of books.

As to the second question, This paper established the AHP model and BP neural network model to predict the score. The three influenced factor in grading is : The number of books, bookmarks, browsing history, the amount of the user's friends , using which we establish evaluation index system. Then, through the establishment of hierarchy analysis model, we get the label number, weight, number of users browsing history friends three index.: 0.0813, 0.6837, 0.2349. And then we determine the score formula on the books of the user. Then 36 groups of data problems are analyzed, and the use of scoring formula is obtained for each book user rating. Then we construct BP neural network model: the original data were chosen for the first 99 sets of datas. We input the number of books, different users label user number of friends, books browsing quantity and the corresponding book score, and we output the score which is from the other users of the books. selecting 80 groups of training data for the neural network, with the remaining 19 sets of data to test the model, making sure the error is within 5.3%. Finally, using the network to predict user the trained on books score. By comparing the two kinds of model so that we can obtain more accurate results

As for question three, we consider that the higher the frequency of the comments from the books , the higher user preference for books, through the users do not read the book selection on the original data of the ID, select the top three of the score for the books five points the highest frequency, which is recommended to the user's three book by ID, then loop five times by five users will eventually get the recommended books ID.

Key words: Books scores; correlation analysis; analytic hierarchy process; BP neural network

Research about recommending books based on Hierarchical analysis method and BP neural network

1. To description of the problem

The development of information technology and the internet has ended an information-shortage era, but has brought the public overloaded information. In the present era, both consumers and producers of information are facing significant challenges: consumers find it is harder to have access to information they are interested in, with explosively growing information around them; and producers also find it is harder to make their information focused by readers.

The conflicts could be solved by a tool called 'Recommendation'. Recommendations are widely used in network products and Apps. Typical styles include the relevant search, topics recommendations, products push in the e-commercial, and 'recommended friends' search in various SNSs.

Attached data are the information of users' behavior at a popular on-line bookstore, including remarks on books, tags of books, and the users' social networks. Consider the following question with the attached data:

1. Analyze the effects on book remarks by users.
2. Design a model to predict the remarks by users on their unread book. Information of users is attached in the file 'predict.txt'.
3. Recommend every user (attached in the file; predict.txt) three books, which they have yet read before.

2. The analysis of the problem

Scores and book recommendation are mainly based on the amount of statistical data processing. Therefore, we need to seize the key and useful data to solve the problem and transform, screen, analyze, summarize the data to analyze the factors affecting the user's score to books, based on which we complete the user score prediction of books and book recommendations. through the establishment of a model of user's score of books.

2.1 Analysis of problem 1

Problem 1 requires us to analyze the factors that affect scores of books. It is a comprehensive analysis of the data in the annex. Firstly, we screen raw data of user_book_score.txt and we get the data of 1-5 points by users; take into account the different factors impact on scores of books, and then filter the data for other analyzes, the relationship between the various stages of books initially been evaluated scores and the number of labels, the relationship with social friends, relations with views of books. Finally, the data obtained were analyzed and summarized the scientific obtain affect users of books score factors.

2.2 Analysis of problem 2

Problem 2 requires to establish a model to predict user score for books in the annex of predict.txt. First, the number of tags, social relationships, books to study three aspects of views, which is a multi-objective decision-making problems. According problem, you can use the software to build hierarchical analysis yaahp total score - two-level model rule layer analysis, using integrated AHP to determine the total score for each indicator weights, and determine a comprehensive evaluation formula books, resulting in books score model for prediction score. Secondly, in order to more fully consider the accuracy of the model, using BP neural network model, first create a neural network structure, the number of different users to tag books, the number of user's friends, books, book views and the corresponding score as input to predict other users of books score as output, based on known data to train the neural network, the process continuously adjust network structure until it reaches satisfied so far, and finally the use of the trained network user score prediction for books.

2.3 Analysis of problem 3

Problem 3 requires us to recommend three unread books for each user according to annex of predict.txt. Considering the book's praise higher the frequency, the higher the level of the user's favorite books, using spss for user_book_score.txt attachment filter, get a fifth of all books rated frequency, and then sort processing in EXCEL, using LOOKUP function screening this user has not read the book to get the ID, select three of the most frequently rated as fifth book is recommended to the user's three books ID, and then finally get circulation problems were analyzed five times the required five users Recommended books ID.

3. The provisions of the symbol

symbols	Explanation
N	the number of Observing samples
y	User ratings for books
x_1	User traffic indicators
x_2	The number of indicators users' Friends
x_3	Tag number of indicators
CR	Consistency proportion

4. The assumption of the question

- 1.Assuming that the impact on the user rating factors are independent of books.
2. Assuming that the only factors affecting the score of books are the books labels, the number of the user's friends and books views.
- 3.Assuming that the frequency of the book praise higher , the user love the book more.

5. The establishment and solution of the model

5.1 The first problem model

5.1.1 Processing of books score data

Topic requires us to analyze the factors that affect scores of books, using SPSS statistical software^[1] Annex user_book_score.txt (User rating data) on scores of books sorted and then filtered to obtain rated 5 stars praise the corresponding books.

5.1.2 Impact of book label on book score

Taking into account the number of books tagged books score results for the impact of the existence of certain relationships, so to conduct a comprehensive analysis on the number of tags. First, using SPSS statistical software accessories book_tag.txt (books tag data) for data filtering, get a different book rated 5 stars appearing several frequency; then counted using EXCEL statistical software, to give the score to 5 points for all books tag number. SPSS software again its bivariate correlation analysis^[2], to obtain descriptive statistics table under the label number factors including mean, standard deviation, the observed number of samples, As shown in Table 1.

Table 1 the number of tags and praise of descriptive statistics

Descriptive Statistics			
	Mean	Standard deviation	N
Tag Number	15.16	7.352	4738
Good Reputation	15.16	7.352	4738

Correlation analysis of the number of tags and ratings, As shown in Table 2.

Table 2 The analysis of the results of label and praise correlation

Correlation			
		Tag	Score
Tag Number	Pearson Correlation	1	0.012
	Significant (Bothside)		0.414
	Square plus cross-products	256030.232	3042.232
	Covariance	54.049	0.642
	N	4738	4738
Favorable comment	Pearson correlation	0.012	1
	Significant (Bothside)	0.414	
	Square plus cross-products	3042.232	256030.232
	Covariance	0.642	54.049
	N	4738	4738

When analyzing Table 2, Pearson correlation coefficient between the number of tags with the books score is 0.012, which means an incomplete correlation and a

positive correlation between the two. Bilateral significant value between unrelated 0.414, denied the two are not related assumptions. Therefore, it can be concluded according to Table 2: There is a positive correlation between the user's score of books and books tag number, the more the number of tags, books, the higher the rating.

5.1.3 Impact of social relationships on the book score

Based on consideration of social relations, to a certain extent the number appears on the user's friends praise fifth the number of books rated frequency, the first use of SPSS software accessories user_social.txt (user's social relational data) for data screening, each user Books rated 5 stars appearing several frequency; then counted using EXCEL statistical software, to give the score to 5 points the number of user's friends. SPSS software using them again bivariate correlation analysis, the number of users affected concern for descriptive statistics Friends scores under the influence of factors including mean, standard deviation, observing the number of samples, As shown in Table 3.

Table 3 the number of users and friends from descriptive

Descriptive Statistics			
	Mean	Standard deviation	N
Tag Number	8.06	6.682	3661
Good Reputation	16.71	29.426	3661

Correlation analysis and scoring of the number of the user's friends are shown in Table 4.

Table 4 Correlation analysis of friends and score results table

Correlation			
		Tag	Score
Number of Friends	Pearson Correlation	1	0.164
	Significant (Bothside)		0
	Square plus cross-products	163411.043	118097.177
	Covariance	44.648	32.267
	N	3661	3661
Favorable comment	Pearson correlation	0.164	1
	Significant (Bothside)	0	
	Square plus cross-products	118097.177	3169049.279
	Covariance	32.267	865.860
	N	3661	3661

When analyzing Table 4, the book ratings and the user number of the user's friends Pearson correlation coefficient was 0.164, which means an incomplete correlation and a positive correlation between the two. Bilateral significant value between unrelated 0, denied the two are not related assumptions. Therefore, it can be concluded according to Table 4: User score between books and book the user's friends there is a positive

correlation between the number, the more the number of the user's friends, the higher the rating the user of books.

5.1.4 Impact of view history on book score

First, we use SPSS statistical software accessories user_read_history.txt (user data read books) to screen data of annex, in this annex different books rated 5 stars appearing several frequency; then use EXCEL statistical software for counting process, get rated as 5 points of views of all the history books. SPSS software using them again bivariate correlation analysis, considering only the history books descriptive statistics table views under this factor include the mean, standard deviation, the observed number of samples, as shown in Table 5.

Table 5 Historical views and praise descriptive statistics table

Descriptive Statistics			
	Mean	Standard deviation	N
Tag Number	579.44	2242.964	4738
Good Reputation	12.91	35.435	4738

Correlation analysis of the history books views and ratings are shown in Table 6.

Table 6 Browsing history correlation analysis and results table

Correlation			
		Tag	Score
By browsing quantity	Pearson correlation	1	0.527
	Significant (Bothside)		0
	Square plus cross-products	23831319985.228	198426991.949
	Covariance	5030888.745	41888.746
	N	4738	4738
Favorable comment	Pearson correlation	0.527	1
	Significant (Bothside)	0	
	Square plus cross-products	198426991.949	5947795.210
	Covariance	41888.746	1255.604
	N	4738	4738

To analyze Table 6, Pearson correlation coefficients views the history of books and books score between 0.527, which means an incomplete correlation and positive correlation between the two. Bilateral significant value between unrelated 0, denied the two are not related assumptions. Therefore, according to Table 6, it can be concluded: the user visits scores of books and books there is a positive correlation, the more traffic history books, books, the higher the rating.

5.1.5 Analysis of the result of overall comprehensive index

According to the solution process, we analyze the factors that affect user's

evaluation from the aspects of books labels, social relations, history, the results were positively correlated with a user score of books. User ratings impact of books were historical views> number of user's friends> number of books tag.

5.2 The second problem model

5.2.1 Hierarchical analysis model

5.2.1.1 The establishment hierarchical structure

According to the problem of a result analysis of data select, the paper will affect the score factors of user of books, the number of labels a book, the history of views, the number of users of the three friends to research, these three factors as the criterion layer using the YAAHO analytic hierarchy diagram software to establish the evaluation score users of books.

As shown in Figure 1.

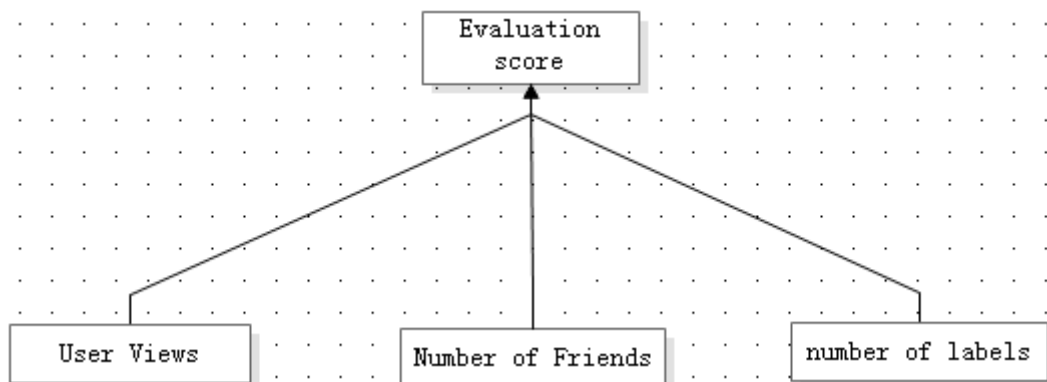


Figure 1 Users of the books score hierarchy diagram

5.2.1.2 Construction of comparative judgment matrix

Comparative judgment 1-9 scale method commonly used, establishing the analysis model of users of the books score level. the total A score goals of books. Book label number A, the number of B users. the user browsing history c paired comparison

matrix. the criterion layer attribute of the target layer judgment.

$$\begin{pmatrix} 1 & 4 & 6 \\ 1/4 & 1 & 4 \\ 1/6 & 1/4 & 1 \end{pmatrix}$$

5.2.1.3 consistency of judgment matrix

In the process of evaluation of some value can not accurately judge. According to the error that exist, will cause the judgment matrix characteristic value deviation, when constructing judgment matrix. Does not require judgment is generally required, otherwise will not analysis. Therefore we must carry on the consistency ratio $CR < 0.1$. That the consistency of consistent judgment matrix. $CR = 0.0932 < 0.1$, the matrix satisfying consistency.

5.2.1.4 To calculate the weight and the establishment of the AHP model

The level of analysis using YAAHP software to get the weight of weight of influence factors on the score of the user books. As shown in table 7.

Table 7 Influence factors on the books score weight table

Alternative	Weights
User Views	0.4232
The number of the user's friends	0.4844
Tag number	0.0924

According to the table 7 influence factors of the formula weights can be user rating for books.

$$y = 0.6837x_1 + 0.2349x_2 + 0.0813x_3$$

5.2.1.5 The results of analysis

The influence of three factors of the user on the books scores were set at 0 to 1 points. that is out of 5 points, to respectively the 3 influence factors, using this formula the users books scores get the find score, thus the user on the books of the score. as shown in table 8.

Table 8 Users on the books of the prediction table

User ID	Book	Prediction score
7245481	794171	5
7245481	381060	4
7245481	776002	5
7245481	980705	4
7245481	354292	5
7245481	738735	5
7625225	473690	4
7625225	929118	4
7625225	235338	5
7625225	424691	5
7625225	916469	4
7625225	793936	4
4156658	175031	4
4156658	422711	5
4156658	585783	5
4156658	412990	4
4156658	134003	4
4156658	443948	5
5997834	346935	2
5997834	144718	2
5997834	827305	2

5997834	219560	2
5997834	242057	2
5997834	803508	2
9214078	310411	3
9214078	727635	3
9214078	724917	3
9214078	325721	3
9214078	105962	3
9214078	235338	3
2515537	900197	3
2515537	680158	3
2515537	770309	3
2515537	424691	4
2515537	573732	4
2515537	210973	3

At this point the AHP model based on the consideration has been successfully established.

5.2.2 BP neural network model

5.2.2.1 The introduction of BP neural network model

BP Neural network belongs to multilayer ANN,a highly nonlinear mapping from input to output.It is made by the input layer and output layer and one or several hidden layers and each layer contains a number of neurons.Between layer and layer of neurons connected by connection weights and . Each layer of neurons state affects only the next layer of neurons state and there is no connections between neurons of the same layer.

BP algorithm is based on the forward propagation and error information.For the input signal,it will spread forth into the hidden layer first,after through the effect function,then spread the hidden layer information to the output layer.If it can not get the desired output in the output layer,it will shift to the error signal back along the original way.Minimizing the error signal by modifying the weights of each neuron.Action function usually adopts S-shaped function.Commonly used the

function $f_{(x)}$ of activation is derivable sigmoid function: $f_{(x)} = \frac{1}{1 + e^{-x}}$

5.2.2.2 The establishment of BP neural network model

The model of Multivariate binomial regression intersection is:

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_m x_m + \sum_{1 \leq j \neq k \leq m} \beta_{jk} x_j x_k$$

In the formula , $\beta_0, \beta_1, \beta_2, \cdots, \beta_m$ are called regression coefficient.

Using the software SPSS to filter the attachment user_book_score.txt first.Through

analysis of 99 pairs of representative sampling data and filter the utilization of the user the number of friends、books tag number、history books'PV.Then we will get the 99 groups of data corresponding to the user the number of friends、books tag number、history books'PV. Building structure of BP neural network model is shown in figure 2.

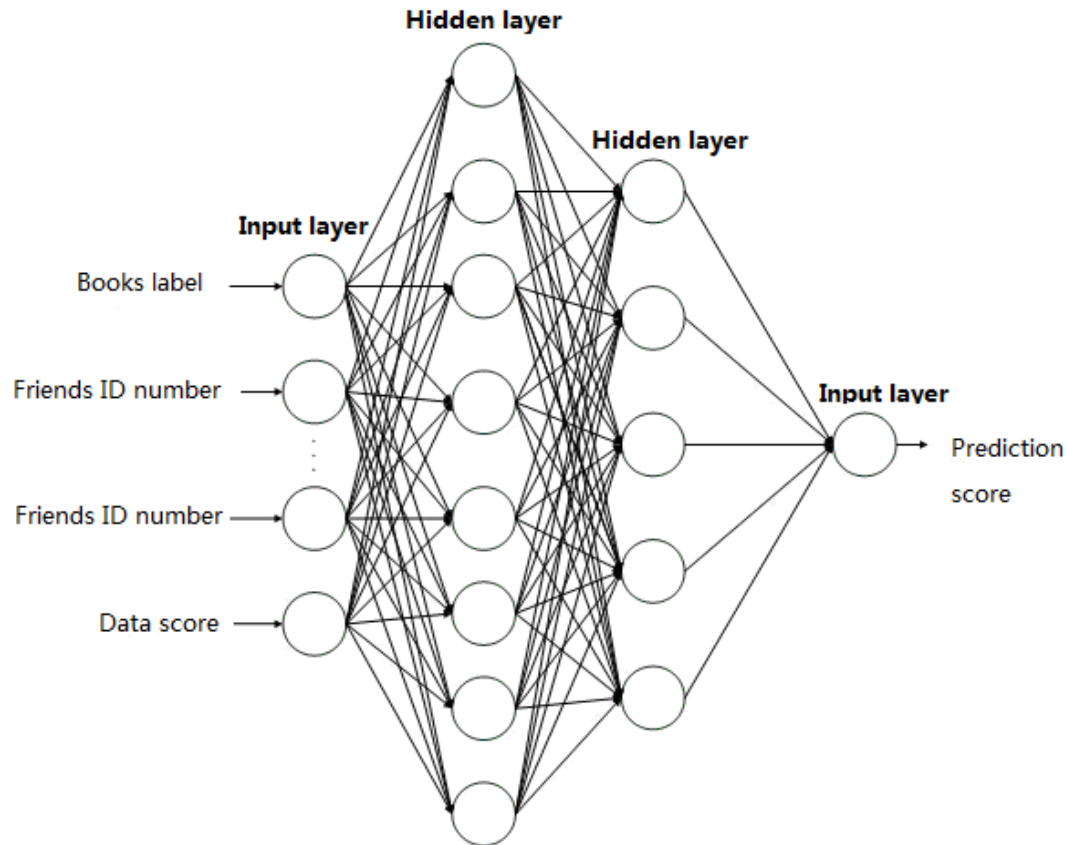


Figure 2 BP neural network structure

Using the selected 99 groups of data,the user the number of friends、books tag number、 history books'PV and books'score these four groups data of the former 80 to train the network.The training use the software MATLAB to implement,on this reason to build user ratings of the BP neural network ^[6]

In order to verify the accuracy of the established BP neural network model,comparing the rest of the 19 sets of data points filtered 99 groups data with the neural network predicted values,in order to test for BP neural network model.Its prediction score and the real score for such as shown in table 9.

Table 9 comparison between predicted score and the true score comparison

User	Book	Score	Prediction score
7245481	284550	4	4
7245481	723581	4	4
7625225	709704	4	4
4156658	430909	5	5
5997834	250272	4	4
9214078	776002	4	4
6249531	545790	3	3

6249531	458404	4	4
7851220	771046	4	5
7851220	837167	3	3
7851220	854951	5	5
9925292	727853	3	3
9925292	728952	4	4
9925228	702699	4	4
9925228	709644	4	4
9912204	920025	3	3
8216735	304843	3	3
8182883	357228	5	5
8182883	357812	4	4

According to table 9 results show that the model error is within 5.3%,Thus the BP neural network model is consistent with the actual score.

At the moment,The network can be used to predict users's corresponding score to books in the attachment predict.txt,then receiving the grading results of books as shown in table 10.

Table 10 User score predicts table of books

User ID	Book	Prediction score
7245481	794171	5
7245481	381060	4
7245481	776002	5
7245481	980705	5
7245481	354292	5
7245481	738735	5
7625225	473690	4
7625225	929118	4
7625225	235338	5
7625225	424691	5
7625225	916469	4
7625225	793936	5
4156658	175031	4
4156658	422711	5
4156658	585783	5
4156658	412990	4
4156658	134003	4
4156658	443948	5
5997834	346935	2
5997834	144718	2
5997834	827305	2
5997834	219560	2
5997834	242057	2
5997834	803508	3

9214078	310411	3
9214078	727635	3
9214078	724917	3
9214078	325721	2
9214078	105962	3
9214078	235338	3
2515537	900197	2
2515537	680158	3
2515537	770309	3
2515537	424691	4
2515537	573732	3
2515537	210973	3

5.2.3 The analysis of model's outcome

According to passage, both Hierarchical analysis model and BP neural network model have been predicted on the basis of the people give a mark to the books, with contributing to two models. Comparing with analysis of the outcomes, we have a result that two models are reasonable models.

5.3 The third problem model

The higher the rate of books praise is, the higher likeability of users to books is. Filtrate attachment of user_book_score.txt by SPSS. Get the rate of the whole books whose score is five and make the sorting process. Then run in the EXCEL. Filtrate and get the books ID which users never look at by LOOKUP function. Select the first highest three books whose the rate of score is five. That is three books ID which recommend to users. Finally get the five users recommendable book ID what the question ask.

Table 11 Recommended Books on the user's results table

User ID	Recommended Books ID
2515537	219029
	144718
	219560
9214078	132378
	105810
	105962
4166658	219029
	144718
	219560
5997834	219029
	144718
	412235
7625225	219029
	144718

6. Evaluation and improvement of the model

In this paper, we establish an analytic hierarchy model to predict the score of books, and all the impact of each factor on the results are quantified, clear and explicit. Based on the three factors that affect the result of the score, we establish a complete set of books scoring system whose prediction is reasonable; The BP neural network model established can automatically extract reasonable rules of output data and input data through learning, and memorize self-adaptive learning content in weights of the network. BP neural network is capable of a high degree of self-learning and adaptation. By learning and training the test sample, and then entering a user ID, books parameters, we get the users' scores. We compare and analyze the results of their score and the results of income AHP to get more accurate results.

In dealing with the factors that affect the score, we take only three factors provided in topic annex into account. We only take into account the number when it comes to book tags. Types of tag can also be analyzed and improved. And when it comes to social relations we take into account only the number of the user's friends. We can collect data by taking into account other factors, optimize and improve the model.

reference

- [1]Chuanhua Yu,SPSS and statistical analysis,Beijing,Publishing House Of Electronics Industry,2007.
- [2]Shengke Cheng,SPSS statistical analysis from entry to the master,Beijing,Tsinghua university press,2010.
- [3] Bingjiang Zhang,AHP and its Applications,Beijing: Publishing House Of Electronics Industry, 2014.
- [4]Wind power prediction problem <http://wenku.baidu.com/view/8fd6ed00eff9aef8941e06e2.html>
- [5]Yan Shi,Li Qun Han,Examples of neural network analysis and design methods,Beijing:Beijing University of Posts and Telecommunications Press,2010
- [6]Xiaochuan Wang,Feng Shi,Yu Lei,MATLAB neural network analysis of 43 cases,Beijing,Beihang University Press,2013.
- [7]Lianfen Wang,Shu Bai Wang,AHP Introduction,Beijing,China Renmin University Press Co. LTD,1989.
- [8]Qiyuan Jiang,Jinxing Xie,Jun Ye,Mathematical Model,Beijing,Higher Education Press,2011.
- [9]Dahai Liu,Ning Li,SPSS15.0 statistical analysis from entry to the master,Beijing,Tsinghua university press,2008.

The appendix

```
BP neural network
function main()
clc
clear all;
close all;
SamNum=99;
TestSamNum=99;
ForcastSamNum=36;
HiddenUnitNum=8;
InDim=3;
OutDim=2;
sqrs=xlsread('last.xls',1,'N1');
sqjdcx=xlsread('last.xls',1,'O1');
sqglmj=xlsread('last.xls',1,'Q1');
glkyl=xlsread('last.xls',1,'M1');
p=[sqrs;sqjdcx;sqglmj];
t=[glkyl];
[SamIn,minp,maxp,tn,mint,maxt]=premnmx(p,t);
rand('state',sum(100*clock))
NoiseVar=0.01;
Noise=NoiseVar*randn(2,SamNum);
SamOut=tn + Noise;

TestSamIn=SamIn;
TestSamOut=SamOut;
MaxEpochs=50000;
lr=0.035;
E0=0.65*10^(-3);
W1=0.5*rand(HiddenUnitNum,InDim)-0.1;
B1=0.5*rand(HiddenUnitNum,1)-0.1;
W2=0.5*rand(OutDim,HiddenUnitNum)-0.1;
B2=0.5*rand(OutDim,1)-0.1;
ErrHistory=[];
for i=1:MaxEpochs

    HiddenOut=logsig(W1*SamIn+repmat(B1,1,SamNum));
    NetworkOut=W2*HiddenOut+repmat(B2,1,SamNum);
    Error=SamOut-NetworkOut;
    SSE=sumsqr(Error) ;
    ErrHistory=[ErrHistory SSE];

    if SSE<E0,break, end
```

```

Delta2=Error;
Delta1=W2'*Delta2.*HiddenOut.*(1-HiddenOut);

dW2=Delta2*HiddenOut';
dB2=Delta2*ones(SamNum,1);

dW1=Delta1*SamIn';
dB1=Delta1*ones(SamNum,1);
W2=W2+lr*dW2;
B2=B2+lr*dB2
W1=W1+lr*dW1;
B1=B1+lr*dB1;
end

HiddenOut=logsig(W1*SamIn+repmat(B1,1,TestSamNum));
NetworkOut=W2*HiddenOut+repmat(B2,1,TestSamNum);
a=postmnmx(NetworkOut, mint, maxt);
x=1990:2009;
newk=a(1,:);
figure ;
subplot(2,1,1);plot(x,newk,'r-o',x,glkyl,'b--+') ;
legend(' Prediction score ',' The original data points');
xlabel('ID');ylabel('book');

pnew=[73.39 75.55
       3.9635 4.0975
       0.9880 1.0268];
pnewn=tramnmx(pnew,minp,maxp);
HiddenOut=logsig(W1*pnewn+repmat(B1,1,ForcastSamNum));
anewn=W2*HiddenOut+repmat(B2,1,ForcastSamNum);
anew=postmnmx(anewn, mint, maxt)

```